

Accounting for Decision Value: Calibration and Selection over Prediction in a Deployed Equity-Forecasting System

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Abstract Most machine-learning work on equity markets is graded on predictive accuracy or area under the ROC curve, and most of it fails to beat a trivial drift baseline once costs and look-ahead are handled honestly. We argue that accuracy measures the wrong thing. What a deployed system delivers is *decision value*: the quality of the small set of positions it actually takes, weighed against the far larger set it stays out of. We make this precise by extending the classical calibration–refinement and reliability–resolution decompositions of forecast quality to the abstaining, cost-aware decision setting, and we prove a tail–AUC decoupling result: a one-sided selective rule can earn a positive edge in its high-conviction tail while global AUC sits at or below one half, so AUC and accuracy systematically misreport what such a system is worth. We then *account* for the value of a real deployed system on the Indian National Stock Exchange across its whole pipeline: multi-source features, a hybrid gradient-boosted/recurrent ensemble, language-model news scoring, post-hoc calibration, a conviction gate, and an append-only audit ledger. The numbers are unforgiving. On 621 resolved forecasts from the live ledger, nominal 90% price intervals covered only 70.7% of outcomes, so deployed confidence was overstated and self-correcting it was nobody’s job. A three-stage recalibration that first repaired a stacking domain mismatch cut expected calibration error from 0.35 to 0.05 without moving accuracy. Single-name next-period direction was unpredictable beyond drift (50.1% out-of-sample, ranking AUC near 0.49). Yet a conviction gate that abstains on roughly 94% of names lifted the 30-day hit rate to 60.6% against a 58.0% base (95% interval 58.8–62.4), positive in four of five test years. Selection produces the realized edge. Calibration is what makes the gate trustworthy enough to act on. The predictive model adds almost no ranking alpha. The practitioner’s lesson follows: instrument the uncertainty layer and report precision at coverage, rather than optimising a number the deployment will not pay for.

Keywords: Decision value, Calibration, Selective prediction, Probabilistic forecasting, Financial machine learning, Model evaluation, Abstention

1 Introduction

A reader who scans the last decade of machine learning applied to stock markets comes away with a puzzle. Paper after paper reports directional accuracies in the high fifties or sixties, areas under the ROC curve comfortably above one half, and backtested Sharpe ratios that would, if real, retire the authors [Sezer *et al.*, 2020; Jiang, 2021]. And yet the people who actually run money are not quietly compounding these edges away in private. The gap between the published record and the deployed reality is wide enough that something in how the field keeps score must be wrong.

We think the something is the scorecard itself. Accuracy and AUC answer a question about *ordering*: given two names, does the model tend to rank the one that rose above the one that fell? That question has a clean answer and a long literature, and it is almost the wrong question for a system that has to decide, name by name, day by day, whether to put on a position or stay flat. A deployed forecaster does not grade every name. It commits to a few and abstains on the rest, and its worth is the quality of the commitments, weighed against the cost of the ones it skipped. Call this *decision value*. Decision value is not accuracy, nor is it a monotone function of AUC. The uncomfortable part is that a system can

have real decision value while its accuracy and AUC report none.

This paper is an exercise in accounting for decision value, carried out on a real system rather than a toy. The system is a production equity-forecasting platform for liquid large capitalisations on the National Stock Exchange of India (NSE), built around the components that have become standard practice: a multi-source feature stack over price, momentum, trend, mean reversion and volatility; a hybrid ensemble that blends gradient-boosted trees with a recurrent sequence model; a language-model layer that scores news; a post-hoc calibration stage; a conviction gate that decides when to act; and an append-only ledger that records every issued forecast so that the system can be graded against the future it actually faced. The platform is live, it has two user-facing surfaces, and since April 2026 it has been writing down what it predicts so that nobody, including its authors, can quietly rewrite the past.

What we found in the live record is worth stating up front, because it sets the agenda for the rest of the paper. First, the deployed uncertainty was wrong in a measurable, one-directional way: nominal 90% price intervals contained the realised price only 70.7% of the time over 621 resolved forecasts, and nothing in the pipeline was responsible for notic-

ing. Second, the model's calibration could be repaired. Expected calibration error fell from 0.35 to 0.05, and the repair added no accuracy; it only pulled the probabilities back onto what the data supported. Third, once the probabilities were trustworthy, the directional skill that the miscalibration had been hiding turned out not to exist: at short horizons, single-name direction across this universe is unpredictable beyond the upward drift of equities, with out-of-sample accuracy at 50.1% and ranking AUC near 0.49. Fourth, and the reason the system is nonetheless useful, a conviction gate that keeps quiet on roughly 94% of names earns a small but repeatable edge on the few it flags: a 60.6% thirty-day hit rate against a 58.0% base, holding up in four of five test years.

Read as accuracy figures these results are a catalogue of failures. Read as a decision-value account they tell a coherent and, we will argue, generalisable story. The predictive model contributes almost none of the realised value. What value there is comes from *selection*, the discipline of acting only in the high-conviction tail, and calibration is what lets the conviction threshold mean what it says. Drift is beta, not skill. The ranking signal that accuracy and AUC are built to reward is, here, absent. Graded on the usual scorecard, the system looks worthless. Graded on the value it can be paid for, it shows a thin but real edge, and you can point to the component responsible.

1.1 Contributions

The paper makes three contributions, one conceptual, one theoretical, and one empirical.

1. *A decision-value accounting framework* (Section 3). We define the decision value of a selective, cost-aware forecaster and decompose it into a drift (base-rate) term, a selection (refinement) term realised at a conviction threshold, and a calibration term that enables rather than adds. The decomposition is the selective-prediction counterpart of the calibration–refinement partition of [DeGroot and Fienberg, 1983] and the reliability–resolution partition of [Murphy, 1973], and it makes “where does the value come from” a question with a measurable answer.
2. *A tail–AUC decoupling result* (Section 3.4). We prove that the selection term is a property of the extreme tail of the score distribution and is largely independent of global ordering: there exist score–label distributions with AUC no greater than one half on which a one-sided conviction rule still earns a strictly positive tail edge. The practical consequence is that AUC and accuracy can report zero skill on a system that has decision value, which is the regime we then measure, and it is why precision at a fixed firing rate is the right instrument.
3. *A whole-system attribution on a deployed platform* (Sections 4–5). We instrument a live system and report measured numbers only: live-ledger interval coverage, a calibration repair that traces to a concrete stacking bug, a verified accuracy ceiling established five ways, and a walk-forward selective signal over eight years. We then carry out the attribution, separating the contribution of drift, calibration, selection and ranking, and showing that ranking contributes essentially nothing while selection carries

the system.

We are careful about scope throughout, in keeping with a standing rule on this project that no reported number is synthetic. Where the live ledger cannot support a claim (most importantly, an isolated return attributable to the news layer, for which per-event labels are not stored), we say so and argue from structure and from the measured end-to-end behaviour instead of manufacturing a clean figure. Section 6 returns to these limits.

2 Related work

Machine learning for markets, and how it is graded. Deep and ensemble models for equity prediction are now routine. Long short-term memory networks [Fischer and Krauss, 2018], gradient-boosted trees and random forests for statistical arbitrage [Krauss *et al.*, 2017], and large cross-sectional studies of machine-learning asset pricing [Gu *et al.*, 2020] are representative, and two systematic reviews map the territory [Sezer *et al.*, 2020; Jiang, 2021]. The dominant evaluation currency in this literature is directional accuracy or AUC, sometimes a backtested Sharpe ratio. A parallel and more sceptical methodological line has shown how fragile those currencies are: backtest overfitting inflates apparent performance [Bailey *et al.*, 2017], the cross-section of “discovered” predictors is riddled with multiple-testing artefacts [Harvey *et al.*, 2016], and a careful treatment of leakage, costs and capacity removes most reported edges [Lopez de Prado, 2018]. Our paper sits inside this sceptical line but turns it constructive: rather than only warn that the usual scorecard misleads, we replace it with a decision-value account and show what the account reveals on a system we operate.

Calibration of probabilistic forecasts. Modern classifiers are frequently overconfident [Guo *et al.*, 2017]. The standard post-hoc remedies are Platt scaling [Platt, 1999] and isotonic regression [Zadrozny and Elkan, 2002], with reliability assessed by the expected calibration error [Naeini *et al.*, 2015] and the Brier score [Brier, 1950; Niculescu-Mizil and Caruana, 2005]. The foundational decompositions of forecast quality are older and, for our purposes, central: Murphy [1973] partitions the Brier score into reliability, resolution and uncertainty, and DeGroot and Fienberg [1983] partition forecaster quality into calibration and refinement. Gneiting *et al.* [2007] crystallise the operating principle these decompositions imply (maximise sharpness subject to calibration), which is the thesis of this paper carried into the decision setting. We use these results as the scaffolding of our framework, not merely as background. A known trap when calibration meets stacked generalisation [Wolpert, 1992] is that a calibrator fitted on one stage's scores and applied to another silently fails. We diagnose this failure in Section 4.3.

Selective prediction and abstention. Allowing a model to decline to answer is an old idea [Chow, 1970] with a modern theory of risk–coverage trade-offs [El-Yaniv and Wiener, 2010; Geifman and El-Yaniv, 2017] and a distribution-free cousin in conformal prediction [Vovk *et al.*, 2005; Angelopoulos and Bates, 2021], including variants built for the non-stationary coverage failures we observe [Gibbs and Candès, 2021]. This community measures risk at a given cov-

erage; the financial-ML community, by contrast, rarely reports whether its probabilities are trustworthy or whether the model knows when to stay out. Our decision-value framework argues that the selective view is the correct one for deployed financial systems, and makes that argument quantitative on a real deployment. The closest framing in spirit treats trustworthy financial AI as an abstention problem; we differ by making the object of study the *attribution* of value across pipeline stages rather than a prescriptive contract, and by supplying the tail-AUC result that explains why the attribution comes out as it does.

Text and returns. That media content carries return-relevant information is well established [Tetlock, 2007; Antweiler and Frank, 2004], that finance-specific dictionaries matter because general ones mislabel domain terms is the lesson of [Loughran and McDonald, 2011], and supervised or pre-trained language models now dominate sentiment extraction [Devlin *et al.*, 2019; Ke *et al.*, 2019]. Whether instruction-tuned models can read financial news and anticipate moves is under active study [Lopez-Lira and Tang, 2023]. Our system uses a language model as one feature source; we are careful, in Section 5, not to claim an isolated alpha for it that our ledger cannot support.

Forecast value and decision-theoretic evaluation. A separate tradition asks not whether a forecast is accurate but whether it is *useful*: what a decision-maker gains by acting on it. Murphy [1993] separated the consistency, quality, and value of a forecast and argued that the three need not move together, an older statement of our complaint that a high AUC need not become a usable signal. Granger and Pesaran [2000] sharpened the point for economics, showing that a statistical loss function and the economic payoff of a decision can rank forecasters in different orders. Proper scoring rules formalise when honest probabilities are rewarded [Gneiting and Raftery, 2007], and in finance the realised worth of a strategy is usually read off a risk-adjusted return such as the Sharpe ratio [Sharpe, 1994]. Decision value belongs to this lineage. It is a value measure, conditioned on a cost-aware action rule, rather than a quality measure averaged over every forecast. What we add is a decomposition of that value into named, separately measurable parts, and a theorem about why the quality measures cannot see it.

Our paper sits at the meeting point of three literatures that rarely talk to each other: the score decompositions of forecast verification, the risk-coverage view of selective prediction, and a financial-ML literature that has learned to distrust its own evaluations. We weld them into an attribution method and run it on a live system.

3 A decision-value accounting framework

This section sets up the bookkeeping. We define the value a selective forecaster delivers, decompose it into interpretable terms, and prove the result that makes the decomposition necessary rather than cosmetic: that the actionable term lives in the tail and is invisible to AUC.

3.1 Forecasts, decisions, and decision value

Fix a horizon of H trading days. For each name and date the system emits a calibrated directional probability $\hat{p} \in [0, 1]$ that the H -day forward return is positive, together with a point forecast and a price interval. Let $y \in \{0, 1\}$ be the realised direction. A *one-sided selective rule* d_τ acts, taking a long position for the horizon, when $\hat{p} \geq \tau$ and abstains otherwise. We focus on the one-sided long/abstain rule because, as Section 5.3 shows empirically, confident short calls on single names are destroyed by drift; nothing in the framework requires one-sidedness.

Two quantities summarise the rule’s behaviour. Its *firing rate* $\phi(\tau) = \Pr[\hat{p} \geq \tau]$ is the fraction of opportunities on which it acts, and its *fired precision* $\pi(\tau) = \Pr[y = 1 \mid \hat{p} \geq \tau]$ is the realised up-rate among the positions it takes. Let $b = \Pr[y = 1]$ be the unconditional up-rate, the base that a name-blind “always act” policy would achieve.

Definition 1 (Selection value and decision value). The *selection value* of the rule d_τ is the edge of its fired positions over the base,

$$\sigma(\tau) = \pi(\tau) - b. \quad (1)$$

Its *decision value* at firing rate ϕ is the selection value scaled by how often the rule is willing to act and net of the per-trade cost κ of acting,

$$V(\tau) = \phi(\tau) [g \sigma(\tau) - \kappa], \quad (2)$$

where g converts a unit of directional edge into a unit of payoff for the position sizing in use.

Equation (2) is intentionally minimal: the realised worth of a selective forecaster is set by how often it acts, how much its acted-upon calls beat the base, and what acting costs. Nothing else enters. Neither accuracy nor AUC appears in it. A model can be excellent on both and still have $V \leq 0$ when its edge does not survive cost; a model at chance on both can have $V > 0$ if it finds a thin, reliable tail. The cost κ and the sizing factor g are venue-specific, so the empirical attribution of Section 5 concentrates on the cost-free core of V (the selection value σ) and returns to the net-of-cost picture once σ has been measured (Section 5.4). The rest of the framework is about where σ comes from and why the usual metrics cannot see it.

3.2 Decomposing fired precision

The fired precision admits a decomposition that mirrors the classical partitions of forecast quality but is conditioned on the act region rather than averaged over all forecasts. Write the act region as $R_\tau = \{\hat{p} \geq \tau\}$ and let $\bar{p}(\tau) = \mathbb{E}[\hat{p} \mid R_\tau]$ be the mean predicted probability there. Adding and subtracting $\bar{p}(\tau)$ in (1),

$$\underbrace{\pi(\tau) - b}_{\text{selection value } \sigma(\tau)} = \underbrace{(\pi(\tau) - \bar{p}(\tau))}_{\text{calibration gap on } R_\tau} + \underbrace{(\bar{p}(\tau) - b)}_{\text{conviction lift}}. \quad (3)$$

The second term, the conviction lift, is how far above the base the model is willing to claim on its acted-upon names; it is pure refinement, the analogue of resolution in [Murphy,

1973]. The first term, the calibration gap, is how wrong that claim turns out to be on those same names; under perfect calibration it is zero and the realised edge equals the claimed lift. Miscalibration enters *only* through this term: a model that systematically over-claims has a negative calibration gap that eats into, or reverses, the lift it advertises. Calibration here enables the edge without adding to its size. It does not generate conviction lift, which is a property of how the model ranks opportunities. But when it is broken, the lift on paper is not the edge you collect, and the threshold τ stops meaning what you set it to.

The decomposition also explains why repairing calibration can leave accuracy untouched while transforming a system's usability. A monotone recalibration map (Platt or isotonic) is order-preserving, so it moves neither accuracy nor AUC nor the conviction lift $\bar{p}(\tau) - b$ evaluated at the matched quantile; a non-monotone calibrator could reorder names and would carry no such guarantee. What the monotone map moves is the calibration gap, pulling $\bar{p}(\tau)$ back onto $\pi(\tau)$ so that the advertised probability equals the delivered one. Section 5.2 shows this happening to a real model.

3.3 Relation to the Brier-score decomposition

The two terms in (3) are the act-region shadows of a classical result. Murphy [1973] split the Brier score of a probabilistic forecaster into three pieces,

$$\text{BS} = \underbrace{\mathbb{E}[(\bar{p}_{\text{bin}} - \bar{y}_{\text{bin}})^2]}_{\text{reliability}} - \underbrace{\mathbb{E}[(\bar{y}_{\text{bin}} - b)^2]}_{\text{resolution}} + \underbrace{b(1 - b)}_{\text{uncertainty}}, \quad (4)$$

where the expectations run over the forecaster's probability bins, $\bar{y}_{\text{bin}}$ is the realised rate in a bin, and b is the base rate. Reliability is small when the forecaster is calibrated; resolution is large when its bins pull outcomes apart; the uncertainty term depends only on the base rate, never on the model. A conviction gate keeps a single bin, the act region R_τ , and reads two of these three quantities in (4) through it. The conviction lift $\bar{p}(\tau) - b$ of (3) is the local face of resolution: a forecaster with no resolution cannot push its acted-on probabilities above the base, so the gate has nothing to fire on. The calibration gap $\pi(\tau) - \bar{p}(\tau)$ is the local face of reliability, and the expected calibration error of (7), defined in Section 4.3, is its size summed over bins. This is the formal reason the attribution falls out the way it does. Selection value is resolution restricted to the tail; calibration is reliability; and the base rate, the uncertainty term, is the drift that neither the model nor the gate may claim. We record the correspondence as a lemma.

Lemma 1 (Local resolution and reliability). *Fix τ and treat the act region R_τ as one bin, with mean forecast $\bar{p}(\tau)$ and realised rate $\pi(\tau)$. The selection value splits as $\sigma(\tau) = (\pi(\tau) - \bar{p}(\tau)) + (\bar{p}(\tau) - b)$, a local reliability gap plus a local resolution lift. If the underlying forecaster has zero resolution, so that its realised positive rate equals the base b on every set fixed by the forecast value, then $\pi(\tau) = b$ for every threshold τ , and therefore $\sigma(\tau) = 0$: no gate can extract an edge.*

Proof. The split is (3) with $\bar{p}(\tau) = \mathbb{E}[\hat{p} \mid R_\tau]$. Zero resolution means the realised rate is b on every event determined

by the forecast value; $R_\tau = \{\hat{p} \geq \tau\}$ is such an event, so $\pi(\tau) = \Pr[y = 1 \mid R_\tau] = b$, whence $\sigma(\tau) = \pi(\tau) - b = 0$. $\square$

Lemma 1 says a gate cannot manufacture an edge from a forecaster that does not separate outcomes. All it can do is harvest whatever resolution exists, which sits in the tail, and only if calibration is good enough that the tail is where the forecaster claims it is. That proviso is the entire job of the calibration repair in Section 5.2.

3.4 Selection lives in the tail: a decoupling from AUC

A decision-value account would just relabel the usual numbers if its central term tracked them. It does not. The selection value can be positive when AUC reports no signal at all. AUC is a global statistic, the probability that a random positive outscore a random negative, and it averages concordance over the whole score distribution. Selection value, by (1), depends only on the conditional up-rate inside a thin top slice. The two can come apart, and on real market data they do.

Proposition 1 (Tail–AUC decoupling). *For every $\varepsilon \in (0, \frac{1}{2})$ and every $\sigma_0 \in (0, 1 - b)$ there is a joint distribution of a score S and label $Y \in \{0, 1\}$ with base rate $\Pr[Y = 1] = b$ such that, for the threshold τ with firing rate $\phi(\tau) = \varepsilon$, the selection value satisfies $\sigma(\tau) \geq \sigma_0$, while the global $\text{AUC}(S, Y) \leq \frac{1}{2}$.*

Proof. We construct such a distribution explicitly. Put all score mass on three cells, ordered $B_1 < B_2 < T$ by score, with ties allowed only within a cell. The top cell T has mass ε and conditional positive rate $b + \sigma_0$ (a valid probability since $\sigma_0 < 1 - b$); the two bulk cells B_1, B_2 each have mass $m = (1 - \varepsilon)/2$ and positive rates $p_1 = b + a$ and $p_2 = b - c$. The threshold $\tau = \inf T$ fires on exactly T , so $\phi(\tau) = \varepsilon$ and $\pi(\tau) = b + \sigma_0$, hence $\sigma(\tau) = \sigma_0$. Holding the overall base at b requires $\varepsilon(b + \sigma_0) + m(p_1 + p_2) = b$, i.e.

$$a - c = -\frac{2\varepsilon\sigma_0}{1 - \varepsilon}. \quad (5)$$

The bulk is *anti-sorted* when $p_1 > p_2$ (equivalently $a + c > 0$): the higher-scored cell B_2 carries the lower positive rate.

Write the AUC as the probability that a random positive outscore a random negative, ties counted as one half, and let $P = b$, $N = 1 - b$ be the total positive and negative mass. Let $(t_1, t_0) = (\varepsilon(b + \sigma_0), 1 - b - \sigma_0)$ be the positive and negative mass of T , and likewise $(u_1, u_0) = m(p_2, 1 - p_2)$ for B_2 and $(v_1, v_0) = m(p_1, 1 - p_1)$ for B_1 . Summing concordance over ordered cell pairs, where a positive–negative pair is concordant when the positive's cell has the higher score, scores one half within a shared cell, and zero otherwise, gives

$$\text{AUC} \cdot PN = \underbrace{t_1(u_0 + v_0) + u_1v_0}_{\text{concordant}} + \frac{1}{2} \underbrace{(t_1t_0 + u_1u_0 + v_1v_0)}_{\text{within-cell ties}}. \quad (6)$$

The discordant pairs (a positive scored below its negative, chiefly the large mass v_1u_0 in which a B_1 positive sits below a B_2 negative) contribute zero, and this is where the

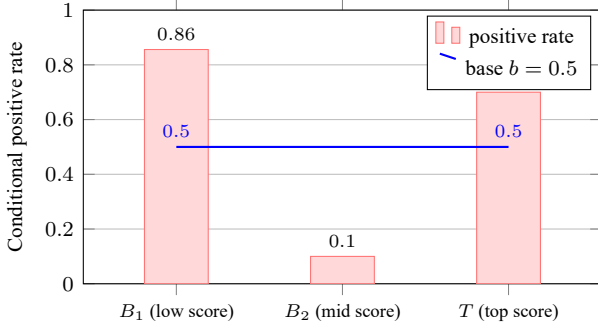


Figure 1. The tail-AUC construction of Proposition 1, drawn for the worked instance ($b = \frac{1}{2}$, $\sigma_0 = 0.2$, $\varepsilon = 0.1$, $c = 0.4$). The bulk is anti-sorted: the low-score cell B_1 carries a high positive rate while the mid-score cell B_2 carries a low one, which drags global AUC down to 0.234. The thin top cell T still sits above the base, so a gate firing only on T banks $\sigma = 0.2$. The figure illustrates the construction and makes no empirical claim.

anti-sorted bulk pays. The two tail terms $t_1(u_0 + v_0)$ are $O(\varepsilon)$. Raising the bulk contrast c , with a following by (5), drives $p_2 \downarrow$ and $p_1 \uparrow$ and lowers the right-hand side of (6); the contrast is feasible up to $p_2 = b - c \geq 0$. For any fixed ε , σ_0 a large enough bulk deficit therefore dominates the $O(\varepsilon)$ tail surplus, giving $\text{AUC} \leq \frac{1}{2}$ while $\sigma(\tau) = \sigma_0$.

A concrete instance settles existence. Take $b = \frac{1}{2}$, $\sigma_0 = 0.2$, $\varepsilon = 0.1$, and $c = 0.4$, so by (5) $a = 0.356$, giving $p_1 = 0.856$, $p_2 = 0.1$, and a top rate $b + \sigma_0 = 0.7$. Then $t_1 = 0.07$, $t_0 = 0.03$; $u_1 = 0.045$, $u_0 = 0.405$; $v_1 = 0.385$, $v_0 = 0.065$, and (6) evaluates to $0.07(0.405+0.065) + 0.045(0.065) + \frac{1}{2}(0.0021+0.0182+0.0250) = 0.0585$, so $\text{AUC} = 0.0585/0.25 = 0.234 \leq \frac{1}{2}$, while the gate earns $\sigma(\tau) = 0.2$ at firing rate $\phi = 0.1$. $\square$

The construction is not a curiosity (Figure 1). A system whose scores are slightly anti-informative in the bulk (common when a model has learned a weak mean-reverting pattern that is wrong on average but concentrates a few genuine signals at the extreme) will report $\text{AUC} \leq \frac{1}{2}$ and look worthless, while a conviction gate reading only its top few per cent collects a real edge. The standard scorecard does not merely understate such a system; it can assign it a failing grade. What does see the edge is precision at a fixed firing rate, $\pi(\tau)$ at $\phi(\tau)$, the quantity our results report and the attribution in Section 5.5 is built on. None of this makes AUC meaningless. For ranking-driven applications it is the right tool. It is the wrong instrument for a selective decision system, and the proposition is why.

3.5 The attribution procedure

The framework yields a procedure that we apply to the deployed system in Section 5. Given a held-out, walk-forward stream of $(\hat{p}, y)$ pairs and the live ledger of resolved intervals, the decision-value attribution is:

1. *Audit deployed uncertainty.* On the live ledger, compare nominal interval coverage to empirical coverage. A gap establishes that confidence is not self-correcting and fixes the stakes (Section 5.1).
2. *Measure the calibration term.* Compute ECE before and after recalibration; verify that accuracy and AUC are un-

changed, isolating the calibration gap of (3) as the only thing that moved (Section 5.2).

3. *Bound the ranking term.* Estimate AUC and accuracy directly to establish whether any global ordering skill exists (Section 5.3).
4. *Measure the selection term.* Tune the conviction threshold τ^* on training folds, then report $\pi(\tau^*)$, $\phi(\tau^*)$ and $\sigma(\tau^*)$ out-of-sample with a confidence interval and a test against the base (Section 5.4).
5. *Settle the attribution.* Express the realised fired precision as $b + \sigma(\tau^*)$ and attribute each component, contrasting it with the ranking-based reading the standard scorecard would give (Section 5.5).

The output is a single accounting that says, in measured numbers, which part of a machine-learning pipeline is doing the paying.

4 The deployed system and the evaluation protocol

The framework is general; the numbers in Section 5 come from one concrete system, which we describe here in enough detail to make the measurements reproducible in spirit. The platform is a live equity decision-support product for the Indian NSE. It runs as two surfaces over a shared model and service layer: a research-facing analytics application and a user-facing web application with a separate forecasting backend. Both read the same trained artefacts, so the numbers in this paper describe the models that real users see.

4.1 Signals, features, and the hybrid ensemble

For each name the system builds a daily feature stack from end-of-day price data: returns over several look-backs, momentum and trend descriptors, mean-reversion indicators, and realised-volatility measures. On top of this it maintains a news layer (Section 4.2) that scores recent headlines for each entity. These features feed a *hybrid ensemble* that blends two model families with complementary inductive biases: a gradient-boosted tree ensemble that captures threshold and interaction structure in the cross-section, and a recurrent sequence model over the price history that captures temporal dependence. The two are combined by a learned blend; the ensemble emits, per name and horizon, a point forecast, a price interval at a nominal confidence level, and a directional probability $\hat{p}$.

We treat the ensemble as a black box for the purposes of this paper. That is not laziness; it is the point. The decision-value account does not care how the probability was produced, only whether it is calibrated and whether the system knows when to trust it. The components that determine those two properties, the calibration stage and the conviction gate, sit downstream of the ensemble, and that is where the analysis concentrates.

4.2 News as a feature: a language model as measurement instrument

The news layer scores each recent headline for an entity and folds the result into the feature stack. Scoring is done by a compact instruction-tuned language model (in the current deployment, Claude Haiku 4.5 from Anthropic), prompted as a quantitative analyst and constrained to emit structured numeric judgements with fixed numerical anchors so that two runs on the same item agree closely. Each scored item is memoised in a content-addressed cache keyed by a hash of the entity and the normalised headline, so a given headline is scored once and reused; when the model is unavailable a transparent lexicon fallback supplies a conservative score that is flagged for downstream consumers. We describe the news layer because it is part of the deployed system and because the journal asks that any use of AI tools be disclosed (see also the declaration at the end of the paper). We do *not*, however, claim a standalone return for it: the live ledger does not store per-event return labels, so an isolated news alpha is not identifiable from our deployment, and we decline to reconstruct one. Its role in this paper is as one input to the ensemble whose end-to-end behaviour we measure.

4.3 Calibration stage

A directional probability is only useful if it is calibrated: among names assigned $\hat{p} \approx 0.7$, about 70% should rise. We measure calibration by the expected calibration error over B equal-width bins,

$$\text{ECE} = \sum_{b=1}^B \frac{|G_b|}{N} |\bar{y}_b - \bar{p}_b|, \quad (7)$$

where $\bar{p}_b$ and $\bar{y}_b$ are the mean predicted probability and mean realised outcome in bin b , complemented by the Brier score $\frac{1}{N} \sum_i (\hat{p}_i - y_i)^2$ [Brier, 1950]. The calibration stage applies a learned monotone map to the ensemble's raw directional score. Section 5.2 documents what happened when we first measured (7) on the deployed model. It was badly broken, and the bug was the system's own.

4.4 The conviction gate

Calibration makes $\hat{p}$ honest; it does not make it skilful. If short-horizon single-name direction is near-random, a calibrated probability sits near the base rate almost everywhere, and acting on every name is pointless. The conviction gate is the one-sided selective rule d_τ of Section 3.1: for horizon H it emits an UP call when $\hat{p} \geq \tau^*$ and NEUTRAL otherwise. The threshold is tuned on training folds as the smallest τ whose fired bucket reaches a target precision p_0 while still firing at least a minimum fraction ϕ_0 of the time,

$$\tau^* = \min \{ \tau : \pi(\tau) \geq p_0 \text{ and } \phi(\tau) \geq \phi_0 \}, \quad (8)$$

with $p_0 = 0.60$ and $\phi_0 = 0.05$ in deployment. The gate is intentionally one-sided for the empirical reason given in Section 5.3. Choosing τ^* on training data and reporting π, ϕ out-of-sample is what keeps the selection-value estimate honest; this is the threshold-tuning analogue of not peeking at the test set.

4.5 The append-only audit ledger

Since April 2026 the deployed system has written every issued forecast to an append-only ledger: entity, issue timestamp, target date, point forecast, interval, nominal confidence, directional probability and horizon. Each row is later stamped with the realised price and outcome on its original target date and is never retro-edited; scoring is first-look only. The ledger matters for two reasons. It is the strongest evidence we have, because it is produced by the running system rather than reconstructed after the fact, and it is the mechanism that makes the system *falsifiable in deployment*: a forecaster that cannot be confronted with the future it actually faced cannot be audited, and an un-auditable forecaster in a money-bearing setting is the thing the backtest-overfitting literature warns about [Bailey *et al.*, 2017].

4.6 Data and evaluation protocol

Two bodies of evidence are kept strictly separate. The first is an *offline walk-forward* study on daily data from 2018 to 2026 over a 54-name large-cap universe, used to fit and evaluate the ensemble, the calibration stage and the gate. All offline statistics are out-of-sample under an expanding window, so no future information enters a fitted fold; the gate threshold is tuned only on training folds. The second is the *live append-only ledger* of Section 4.5. We report, from the offline study, the calibration repair, the accuracy ceiling and the selective signal; from the live ledger, interval coverage. No quantity in Section 5 is illustrative unless its caption says so. Figure 2 places the components in one pipeline and marks where each measurement is taken.

Two further points bear on how far the numbers can be trusted. Both deployment surfaces of Section 4 read the same trained artefacts, so the figures below describe what real users see rather than a research-only variant. And the offline study is reproducible in the sense an audit cares about. The universe is fixed in advance; the walk-forward windows expand rather than slide, so no fitted fold sees its own future; the gate threshold τ^* is selected on training folds alone; and every persisted statistic traces to a stored artefact rather than to a number typed into the manuscript. The live ledger adds the stronger guarantee that the running system, not a reconstruction, produced each forecast, and that the score against the outcome is taken on the original target date and never revised afterwards.

5 Results: an account of where the value comes from

We now run the attribution procedure of Section 3.5 on the deployed system, in order.

5.1 Deployed uncertainty does not self-correct

We start with the live ledger because it sets the stakes. Between 19 April and 12 June 2026 the system issued and later resolved 621 price-interval forecasts on seven NSE

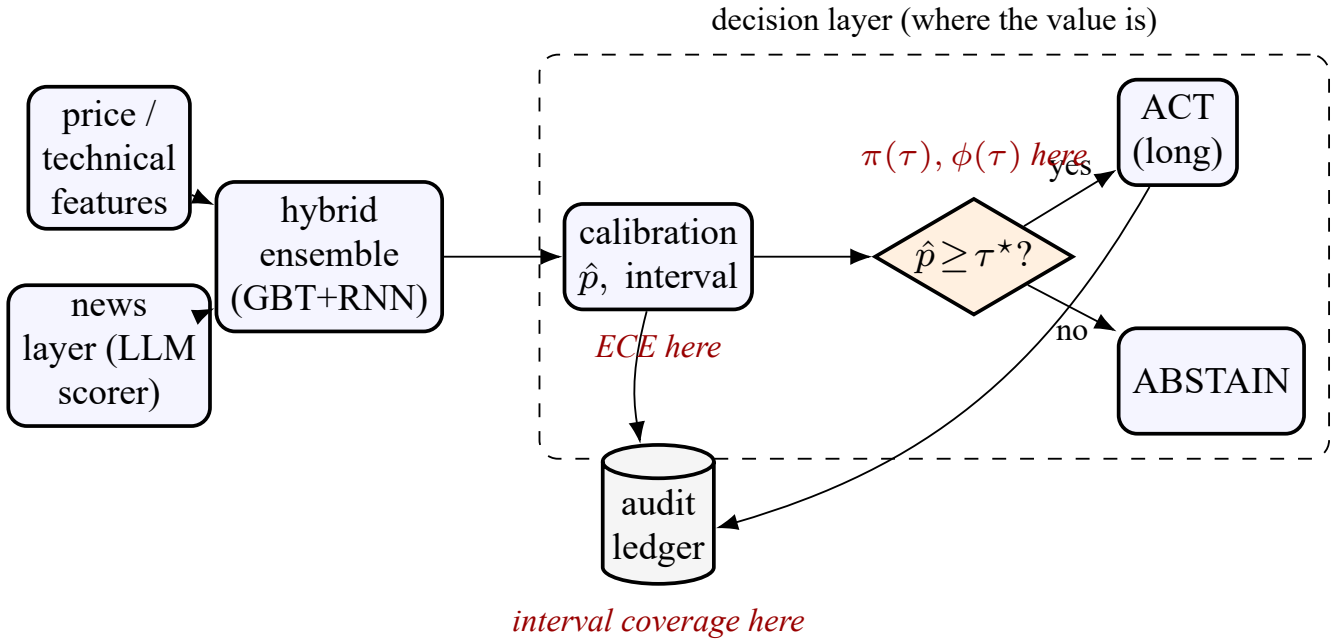


Figure 2. The deployed pipeline and where each measurement in Section 5 is taken. Features (price and news) feed a hybrid ensemble; its directional probability is calibrated, then passed to a conviction gate that acts or abstains; every issued forecast is written to an append-only ledger and later scored against its target date. The dashed box is the decision layer the attribution finds responsible for the realised value. The schematic is conceptual.

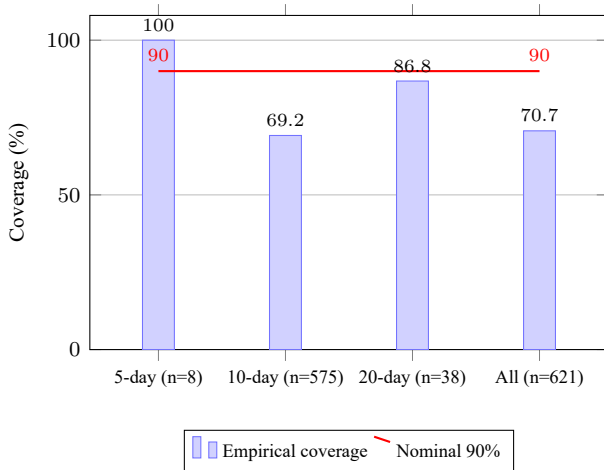


Figure 3. Empirical coverage of nominal 90% price intervals on the live ledger, by horizon. The deployed intervals are materially under-covered at the dominant ten-day horizon, which carries 575 of the 621 forecasts.

large caps (ABB, HDFCBANK, ICICIBANK, INFY, RELIANCE, SBIN, TCS), each carrying a nominal 90% interval with a median half-width of 5.6% of price. Were the intervals well calibrated, about 90% should contain the realised price. They did not: empirical coverage was 70.7% overall, a 19.3 percentage-point shortfall. Most forecasts were at the ten-day horizon, where coverage was 69.2% over 575 resolved forecasts; the smaller twenty-day and five-day buckets covered 86.8% and 100% (Table 1, Figure 3). Put plainly, realised prices broke through the $\pm 5.6\%$ band far more often than one time in ten, so the deployed model’s stated confidence was systematically overstated, and nothing in the pipeline noticed on its own. This is the empirical reason calibration and abstention cannot be afterthoughts, and it is step one of the attribution: before any value can be claimed, the uncertainty has to be shown to be untrustworthy as shipped.

5.2 The calibration term: a stacking bug, and a three-stage repair

The offline diagnosis of overconfidence is sharper still; this and the two results that follow are walk-forward measurements, not live-ledger figures. The deployed model’s next-day directional head reported probabilities clustered near 0.80 against a realised accuracy of about 0.51, giving $ECE = 0.345$ and a Brier score above 0.366, worse than always predicting 0.5. Tracing the pipeline exposed a textbook stacked-generalisation trap [Wolpert, 1992]: an isotonic calibrator had been fitted on the out-of-fold *average of the base learners* but then applied to the *stacked* prediction, a different distribution, and the final blend carried no calibration of its own. The fix is mechanical. We refit the isotonic map on the same base-average it was meant to correct, form the blend, then fit a final Platt scaler [Platt, 1999] on a held-out fold and apply it to both the test rows and the live row. This cut ECE from 0.345 to 0.049 and pulled the reported up-probability back to a realistic neighbourhood of 0.50.

The crucial observation, and the reason this is step two of the attribution rather than a contribution in its own right, is that *accuracy did not move*. The recalibration map is monotone, so by the argument of Section 3.2 it cannot change the model’s ordering of names, its accuracy, its AUC or its conviction lift. It changed only the calibration gap, pulling the advertised probability onto the delivered one. In decision-value terms, calibration did not create value; it made the value that selection would later extract *collectible*, by ensuring that the threshold τ^* would mean what it was set to mean. A system shipped at $ECE = 0.345$ would have a conviction gate firing on a probability scale that lies by 30 points; the gate would be tuned on a fiction. The episode is also a clean, transferable lesson: a calibrator is bound to the distribution it was fitted on, and stacking quietly creates more than one

Table 1. Live-ledger interval coverage by horizon (NSE large caps, 19 April–12 June 2026). All intervals carry a nominal 90% level.

Horizon	Resolved forecasts	Nominal coverage	Empirical coverage
5 trading days	8	90.0%	100.0%
10 trading days	575	90.0%	69.2%
20 trading days	38	90.0%	86.8%
All	621	90.0%	70.7%

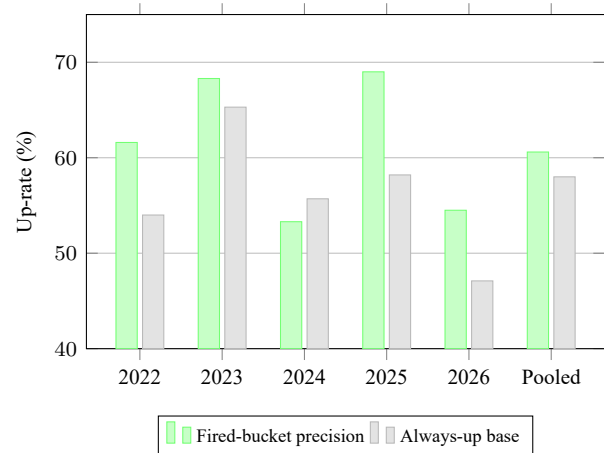
such distribution.

5.3 The ranking term is essentially zero

Once calibration was repaired, it exposed an uncomfortable fact that the inflated probabilities had hidden. With $\hat{p}$ pulled back to the base rate, we asked directly whether any robust single-name directional edge survives at short horizons. A cross-sectional model predicting five-day direction across the universe reached 50.09% out-of-fold accuracy on 131,809 ticker-days, with ranking AUC near 0.49, which is chance. We confirmed the result five ways, varying the learner and the feature set and gating by trend and by market regime: a per-ticker ensemble, a cross-sectional logistic model, a gradient-boosted model with reversion features, trend/momentum gating, and market-regime gating. None produced a stable edge over the always-up base rate at the daily scale; in this sample the trend-gated variants sat *below* the base rate, the universe being mildly mean-reverting at that horizon. This is consistent with the efficient-market prediction for liquid names [Fama, 1970; Malkiel, 2003], and it is step three of the attribution: the ranking term that accuracy and AUC are built to reward is, here, absent. By Proposition 1, that absence does not by itself condemn the system (an edge can still live in the tail), but it does mean that any value we find must be a tail property, and cannot be a ranking one.

5.4 The selection term carries a small, stable edge

Lengthening the horizon to thirty trading days (about a month and a half, the use case the product is actually sold for) opens two modest edges: the upward drift of equities, since about 58% of thirty-day windows are up, and a small idiosyncratic component that a calibrated cross-sectional model can find on momentum, trend, reversal and volatility features. Walk-forward from 2018 to 2026 over the 54-name universe (91,471 training rows, 50,272 out-of-sample rows), the conviction gate of (8) settles at $\tau^* = 0.63$. Its fired UP bucket achieves 60.6% directional precision against a 58.0% always-up base,¹ a selection value of $\sigma(\tau^*) = +2.6$ percentage points, while firing on only 5.65% of observations; the gate abstains on the remaining 94%. The fired bucket pools to 2,840 calls, so the 60.6% precision carries a 95% binomial

**Figure 4.** Selective signal versus its base rate by test year. The conviction gate beats the always-up base in four of five years and pooled, while firing on about 5.6% of observations.

interval of 58.8–62.4%; a one-sided test against the 58.0% base gives $z \approx 2.8$ ($p \approx 0.003$), so the pooled edge is distinguishable from the base at conventional levels. Whether this clears cost is the net-of-cost question of (2). We lack position-level fills to compute V exactly, but the order of magnitude is reassuring: at a thirty-day horizon the typical absolute move in these names runs to several per cent, against a round-trip cost of order 0.2% for liquid large caps, so a long/abstain edge of 2.6 points is not erased by friction. A precise V awaits position-level accounting, which we flag among the limitations.

We do not over-read it. The per-year fired counts are small (Table 2), so the year-to-year swings, from +10.8 points in 2025 to −2.4 in 2024, are within sampling noise, and the fair summary is a small but repeatable tail edge rather than a strong one. The gate beats its base rate in four of the five test years (Figure 4); the single exception, 2024, sits in the table with the rest. The ranking AUC of this signal is about 0.47, at or below chance, which is the regime Proposition 1 describes: the edge lives in the high-conviction tail and not in the ordering, so precision at a fixed firing rate registers it while AUC overlooks it. Confident DOWN calls, for completeness, reach only about 10% precision out-of-sample, destroyed by drift, which is why the gate is one-sided and never emits them.

5.5 Settling the attribution

We can now state, in measured numbers, where the realised decision value comes from. The system's only positive, collectible edge is the fired precision of its conviction gate, $\pi(\tau^*) = 60.6\%$. By the decomposition (3) and Definition 1 this splits cleanly into the components the four preceding sub-

¹The pooled base of 58.0% is the unconditional out-of-sample up-rate over all 50,272 rows (the always-act comparator b of Definition 1), not the fires-weighted mean of the per-year bases in Table 2, which is 55.9%. The unconditional rate is the more conservative comparator; measured against the fires-weighted base the pooled edge would read +4.7 points instead of +2.6.

Table 2. Walk-forward selective signal, by test year. Precision is the realised up-rate of the fired high-conviction bucket; base is the always-up rate that year; fires is the number of high-conviction calls that year.

Test year	Fires	Fired-bucket precision	Always-up base	Edge (pp)
2022	365	61.6%	54.0%	+7.6
2023	290	68.3%	65.3%	+3.0
2024	932	53.3%	55.7%	−2.4
2025	822	69.0%	58.2%	+10.8
2026	431	54.5%	47.1%	+7.4
Pooled	2,840	60.6%	58.0%	+2.6

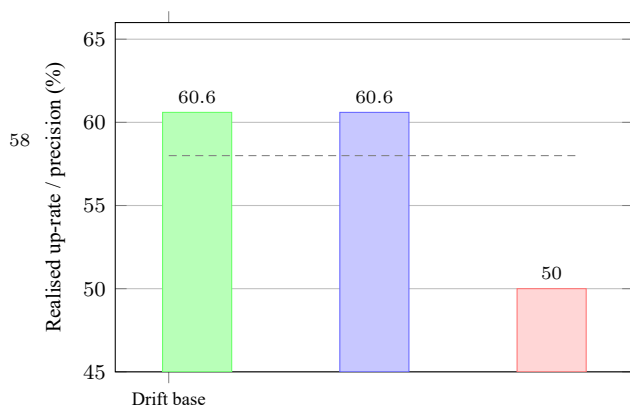


Figure 5. Decision-value waterfall. The drift base (58.0%) plus the selection value of the conviction gate (+2.6 pp) account for the realised fired precision (60.6%). The two right-hand bars show what the conventional scorecard reports for the same signal, and are not up-rates: directional accuracy at chance (50.0%) and a ranking AUC of about 0.47 (plotted as $\text{AUC} \times 100$ for comparison), neither of which sees the tail edge. The dashed line marks the drift base. All values are measured (Sections 5.3–5.4).

sections measured, summarised in Table 3 and drawn as a waterfall in Figure 5.

The split is clean. Of the 60.6 points of realised fired precision, 58.0 are drift (a market beta any index fund collects), and 2.6 are the selection value the system adds. The ranking term that accuracy (50.0%) and AUC (≈ 0.47) measure contributes nothing; on the standard scorecard this signal is a failure, and a practitioner who graded it that way would discard a small but statistically distinguishable edge. Calibration appears nowhere as an additive term, just as Section 3.2 predicts, and yet without the repair from $\text{ECE} = 0.345$ to 0.049 the conviction threshold would have been set on a lying probability scale and the 2.6-point edge would not have been collectible. So the account says three things at once. The model’s prediction is not where the value sits. Selection is. And calibration is the precondition that lets the selection edge be banked. Table 3 captures the economics of this system, and the pattern, we argue next, is not peculiar to it.

6 Discussion

6.1 What the attribution generalises to

The numbers are from one platform; the shape of the attribution is not. Any deployed forecaster that decides whether to act, in a domain where the base rate is high and the ordering signal is weak, will show a similar profile: a large beta-like base, a thin selection edge in the tail, a calibration

term that enables rather than adds, and a ranking term the standard scorecard over-weights. Markets are the cleanest example because their efficiency drives the ranking term toward zero [Fama, 1970; Malkiel, 2003] and their outcomes are public and unforgiving, but the structure recurs wherever a model triages a stream of opportunities and is judged on the few it picks. The recommendation that follows is concrete and portable, and we set it out as a reporting checklist in Section 6.4. A great deal of financial-ML work optimises the ranking term, which our results suggest is the one term least likely to survive deployment.

6.2 Why the standard scorecard misleads, restated

Proposition 1 carries the generalisation. Accuracy and AUC summarise global behaviour. A selective system is defined by what happens in a thin tail. The two are provably decoupled. This is not a quirk of our data. It is why a literature that reports AUC can simultaneously be full of models that look skilful and empty of models that deploy: the metric rewards a property (global ordering) that is both easy to overfit in backtests [Bailey *et al.*, 2017; Harvey *et al.*, 2016] and largely irrelevant to a system that acts on its tail. Reporting precision at coverage, as the selective-prediction community does [Geifman and El-Yaniv, 2017; El-Yaniv and Wiener, 2010], aligns the metric with the deployment. The decision-value account grades a model on the trades it would actually make.

6.3 Threats to validity and limits

Five limits deserve emphasis. First, the live ledger is young, just two months and seven names, so its 19.3-point coverage shortfall, though large and consistent at the dominant horizon, is a current snapshot rather than a stationary law, and the directional-probability sample on the ledger is still too small (28 resolved next-period calls) to support a live reliability diagram; our live claim therefore rests on the 621-forecast interval-coverage evidence, and the calibration and selection results are offline walk-forward measurements. Second, the selection edge is small in absolute terms (2.6 points) and its per-year variation is within sampling noise; we report it as a thin, repeatable tail edge and explicitly not as a strong signal. Third, the news layer’s contribution is argued from its place in the pipeline and not from an isolated backtest, because the deployment does not store per-event return labels; we prefer to say this plainly than to reconstruct a clean number, which is the kind of exercise the backtest-overfitting cri-

Table 3. Decision-value attribution of the deployed system. Each row is a measured quantity from Sections 5.1–5.4; the realised fired precision is the sum of the drift base and the selection value, with ranking and calibration acting as described.

Component	Measured value	Role in decision value
Drift (base rate b)	58.0% up-rate	Beta, not skill; available with no model.
Selection value $\sigma(\tau^*)$	+2.6 pp	The only positive, collectible edge; a tail property.
Ranking skill ($\text{AUC} - \frac{1}{2}$)	−0.01 to −0.03	Daily ≈ -0.01 , 30-day ≈ -0.03 ; invisible to selection (Prop. 1).
Calibration (ECE)	0.345 $\rightarrow$ 0.049	Enabling, not additive; makes τ^* trustworthy.
Deployed interval coverage	70.7% vs 90% nominal	The risk the attribution exists to expose.
Realised fired precision $\pi(\tau^*)$	60.6% = $b + \sigma$	What the system can actually be paid for.

tique exists to catch. Fourth, the calibration drifts over time and must be maintained on a schedule; we treat the maintenance question as out of scope here and a matter for separate work, and make no claim about an optimal recalibration cadence in this paper. Fifth, we attribute the cost-free selection value σ rather than the full net-of-cost decision value V of (2); the order-of-magnitude argument of Section 5.4 suggests the edge survives friction, but a position-level computation of V from realised fills and sizing is left for future work.

6.4 Reporting decision value: a short checklist

The attribution is easy to adopt, so we state it as a checklist a practitioner can run against any deployed selective forecaster. Publish a coverage audit first: take the live ledger, compare nominal with empirical interval coverage, and report the gap by horizon, as in Table 1. Report calibration before and after any post-hoc fix, with the expected calibration error and a note that accuracy did not move, which certifies the fix touched only reliability. State the base rate b explicitly and compare the fired bucket against it, not against an even-odds prior; the drift in a long-only equity setting is large, and it is not skill. Report precision at a fixed firing rate, $\pi(\tau)$ at $\phi(\tau)$, with a confidence interval and a test against b , in place of accuracy or AUC. Separate the ranking term: give the AUC, and if it sits near or below one half while tail precision is positive, say so, because that pattern is the signature of a tail-only edge. Finally, where costs are known, convert the selection value to a net-of-cost figure through (2); where they are not, scope the claim to the cost-free core and name the quantity being reported. A paper that fills in these six lines is hard to misread, and a system that cannot fill them in is not yet ready to be trusted with a position.

6.5 Relation to selective and conformal prediction

The conviction gate is an instance of selective classification [Chow, 1970; Geifman and El-Yaniv, 2017]; what the decision-value account adds is the attribution, separating the

gate’s selection value from the drift it rides on and the calibration that makes it bankable, together with the tail–AUC result that justifies measuring it at a fixed firing rate. The interval under-coverage we documented in Section 5.1 is the failure that distribution-free methods are built to address; adaptive conformal inference [Gibbs and Candès, 2021; Angelopoulos and Bates, 2021] is designed for this kind of non-stationary coverage shortfall, and replacing the deployed parametric intervals with an adaptive conformal layer is the clearest next step the attribution points to.

7 Conclusion

Machine-learning systems for markets are usually graded on accuracy and AUC, and most of them, graded carefully, have neither. We have argued that this is the wrong way to keep score. A deployed forecaster is paid for the quality of the positions it actually takes, weighed against those it skips, and we made that quantity, its decision value, precise by extending the calibration–refinement and reliability–resolution decompositions of forecast verification into the abstaining, cost-aware decision setting. The framework comes with a theorem that explains why the usual scorecard cannot see what such a system is worth: selection value lives in the tail of the score distribution and is provably decoupled from global AUC, so a system with real decision value can be reported at chance. We then ran the attribution on a live equity platform. The result is uncomfortable, and we believe it is representative. Deployed 90% intervals covered only 70.7% of outcomes, so confidence had to be earned rather than assumed; a calibration repair cut expected calibration error from 0.35 to 0.05 without adding a point of accuracy; single-name direction proved unpredictable beyond drift; and a conviction gate that abstains on 94% of names nonetheless banked a 60.6% hit rate against a 58.0% base, in four of five years. Drift is just beta. The edge comes from selection. Calibration is the precondition that makes it bankable. And prediction, the thing the field optimises for, contributes almost nothing. The practical message follows: instrument the uncertainty layer, grade the model on the trades

it would make, and put the engineering effort where the attribution says the value is. Replacing the deployed intervals with an adaptive conformal layer, and a per-event labelled evaluation of the news contribution, are the clear next steps.

Declarations

Use of generative AI and AI-assisted technologies

Two distinct uses are disclosed. (i) *As an instrument within the system under study*: an instruction-tuned large language model (Claude Haiku 4.5, Anthropic) scores news headlines for the feature layer described in Section 4.2, from the anchored prompt summarised there; these outputs are part of the deployed pipeline, were validated against a deterministic lexicon fallback, and are persisted for audit. (ii) *In manuscript preparation*: during the writing of this paper the authors used a large language model solely to improve the language and readability of author-written drafts. No generative tool produced data, results, figures or interpretations; every quantitative finding derives from the authors' deployed system and its measured ledger. After using these tools the authors reviewed and edited the content and take full responsibility for the publication.

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Authors' Contributions

AP conceived the decision-value accounting framework, built and operated the deployed system, performed the experiments, and drafted the manuscript. SD supervised the work, contributed to the evaluation design and the analysis, and reviewed and edited the manuscript. Both authors read and approved the final manuscript. Author contributions follow the CRediT taxonomy (<https://credit.niso.org/>).

Competing interests

The authors declare that they have no competing interests.

Availability of data and materials

The live forecast ledger and the offline walk-forward statistics underlying Tables 1–3 and Figures 3–5 can be made available by the corresponding author on reasonable request, subject to the platform's terms. Source price data are public NSE end-of-day records.

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